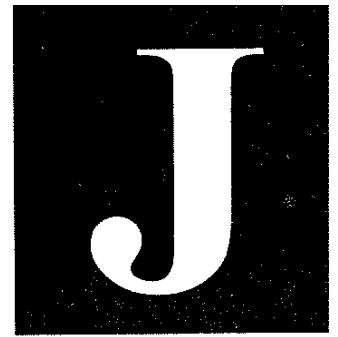


THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Bhaskara Contest

(NMTTC- at **JUNIOR LEVEL** IX & X Standards)

Saturday, 23rd August 2014.



Note:

- 1) Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.
- 2) Diagrams are only visual aids; they are not drawn to scale.
- 3) You are free to do rough work on separate sheets.
- 4) Duration of the test: 2 p.m. to 4. p.m.- 2 hours.

PART – A

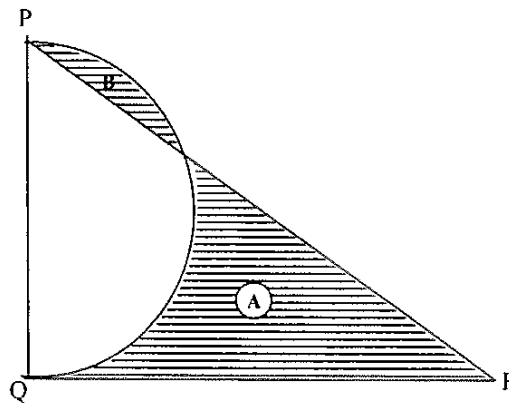
Note:

- Only one of the choices A,B,C,D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

1. For some natural number ' n ', the sum of the first ' n ' natural numbers is 240 less than the sum of the first $(n + 5)$ natural numbers. Then n itself is the sum of how many natural numbers starting with 1.

a) 7 b) 6 c) 9 d) 10

2. In the adjoining figure, $PQ = 42$ cm. QR is the tangent to the semicircle at Q . If the difference of the areas of regions A , B is 357 cm^2 , then the base QR of the right triangle PQR is (in cm)
(Take $\pi = 22/7$)



a) 42 b) 48 c) 52 d) 50

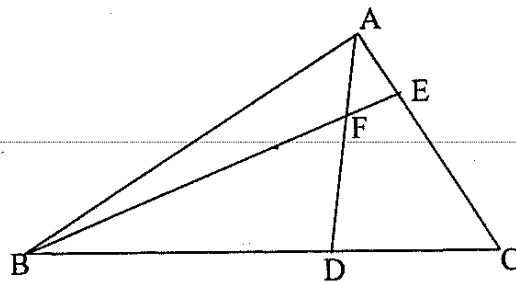
3. By rearranging the digits of the integer 1288, we get a total of 12 different integers including 1288. The sum of all these twelve integers is

a) 577727 b) 63327 c) 72227 d) 466627

4. The sides of a right angled triangle are all integers. Two sides are primes that differ by 50. The smallest possible value of the third side is

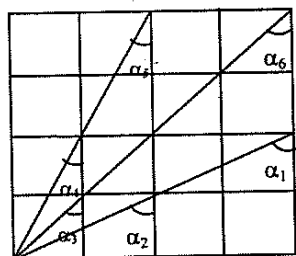
a) 60 b) 57 c) 53 d) 49

5. In $\triangle ABC$, D is on BC such that $BD:DC = 3:1$. E is on AC such that $AE:EC = 2:3$. AD , BE cut at F
 Area of $\triangle AFE = 4 \text{ cm}^2$
 Area of $\triangle BFD = 30 \text{ cm}^2$
 Area of the triangle ABC (in cm^2) is



- a) $\frac{185}{3}$ b) $\frac{225}{3}$ c) $\frac{241}{3}$ d) $\frac{230}{3}$

6.



Observe the angles $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$, and α_6 in the square and shown. The measure of grid $(\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6)$ is

- a) 180° b) 270° c) 360° d) 225°

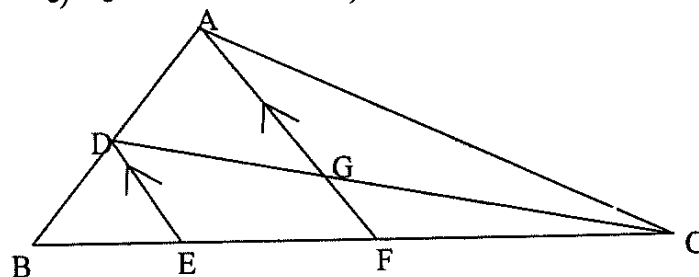
7. N is a five digit number. 1 is written after the 5 digits of N to make it a six digit number, which is three times the same number with 1 written before N . (If $N = 23456$ it means 234561 and 123456). Then the middle digit of the number N is

- a) 2 b) 4 c) 6 d) 8

8. The five digit number $a679b$ is a multiple of 72. Then the value of $a + b$ is

- a) 3 b) 5 c) 6 d) 7

9. In the adjoining diagram D is the midpoint of AB . A line DE is drawn to cut BC at E . AF is parallel to DE . It is given that $EF = FC$, $AF = 10 \text{ cm}$ and area of $\triangle BDE$ is 5 cm^2 , the area of $\triangle AGD$ (in cm^2) is



- a) 8 b) 8.5 c) 7.5 d) 7

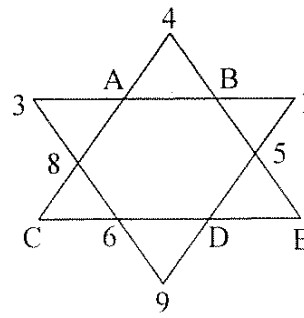
10. The number of two digit numbers having the property that when they are divided by the sum of their digits, the quotient is 7 without remainder is

- a) 0 b) 1 c) 3 d) 4

11. The sum of all values of integers n for which $\frac{n^2 - 9}{n - 1}$ is also an integer is

- a) 0 b) 7 c) 8 d) 9

12. The numbers 1 to 12 are placed in the figure as shown. The sums of the numbers in each of the 6 rows are the same. The number 7 must go to the place marked



- a) A b) B c) E d) D
13. There are 2014 people sitting around a big round table dinner. Each person shakes hands with everybody except the persons sitting on both sides of him. The total number of handshakes that takes place is
- a) 1007×2014 b) 2014×2012 c) 1007×2011 d) 1007×2012
14. A , B and C run for a race which is a straight road of x meters. A beats B by 30 meters B beats C by 20 metres, A beats C by 48 metres. Then x (in meters) is
- a) 150 b) 200 c) 300 d) 500
15. In triangle ABC , $\angle B = 2\angle C$. AD is the angle bisector such that $DC = AB$. Then the measure of $\angle A$ is
- a) 60° b) 72° c) 84° d) 108°

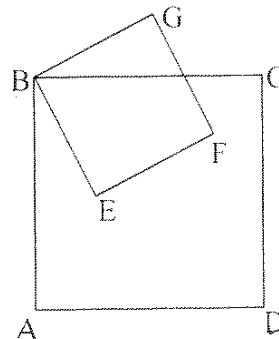
PART – B

Note:

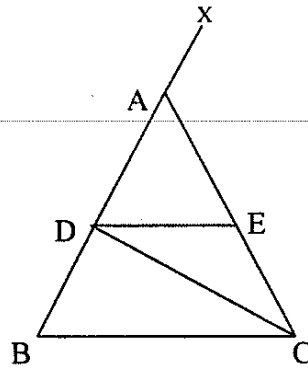
- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

16. The six digit number that increases 6 times when its last three digits are carried to the beginning of the number without their order being changed is _____
17. If a , b , c are real and $a + b + c = 0$ the value of $a(b-c)^3 + b(c-a)^3 + c(a-b)^3$ is _____
18. The population of a town is 20000. The annual birth rate is 4 % and the annual death rate is 2%. The population of the town after 2 years is _____
19. The symbol $\lfloor x \rfloor$ means the integral part of x .
[Ex: $\lfloor 2.3 \rfloor = 2$, $\lfloor \sqrt{15} \rfloor = 3$]. Then the value of $\lfloor \sqrt{1} \rfloor + \lfloor \sqrt{2} \rfloor + \lfloor \sqrt{3} \rfloor + \dots + \lfloor \sqrt{99} \rfloor + \lfloor \sqrt{100} \rfloor$ is _____

20. $ABCD$ is a square $BEFG$ is another square drawn with the common vertex B such that E , F fall inside the square $ABCD$. Then $DF = \frac{1}{n} AE$ where n is



21. In the adjoining figure $AB = AC$. The exterior angle $CAX = 140^\circ$. D is the point on AB such that $CB = CD$. DE is drawn parallel to BC to meet AC at E . The measure of the $\angle DCE$ is _____



22. m, n are natural numbers. If $(m-8)(m-10) = 2^n$, the number of pairs (m, n) is _____

23. If $f(x) = \log\left(\frac{1+x}{1-x}\right)$ for $-1 < x < 1$ and it is found that $f\left(\frac{3x+x^3}{1+3x^2}\right) = K f(x)$, then the value of K is _____

24. If a, b, c, d are positive integers such that $a^5 = b^4$, $c^3 = d^2$ and $c-a = 19$, then the numerical value of $d-b$ is _____ (you can express in powers of numbers)

25. The contents of two vessels containing water and milk in the ratio 1: 2 and 2: 5 are mixed in the ratio 1: 4. The resulting mixture will have water and milk in the ratio _____

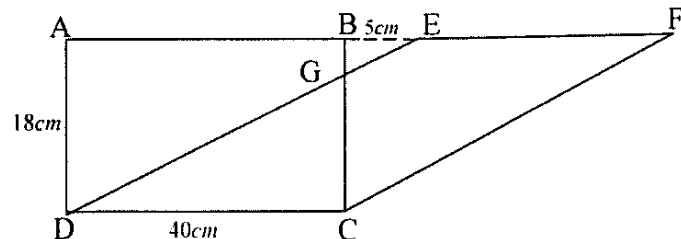
26. $n = 560560560560563$. Saket divided n^2 by 8. He will get a remainder _____

27. The least positive integer by which 396 be multiplied to make a perfect cube is _____

28. The value of $\sqrt[3]{\frac{1.2.4 + 2.4.8 + \dots + n.2n.4n}{1.3.9 + 2.6.18 + \dots + n.3n.9n}}$ is _____

29. n is a natural number. It is given that $(n+20) + (n+21) + \dots + (n+100)$ is a perfect square. Then the least value of n is _____

30. $ABCD$ is a rectangle $DEFC$ is a parallelogram. $ABEF$ is a straight line. Area of the quadrilateral $CGEF$ is _____



_____()